

Unifying Special Relativity and Quantum Mechanics

Lecture 16 (Addendum) — The Weinberg Soft Theorem for Photons

(Soft emission from external legs, the Ward identity, charge conservation, and the closing of the circle)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–17: everything in the normal text and in the **yellow / pink / grey** boxes is a faithful transcription of the handwritten notes; the colours of the text roughly follow the pens of the original. Everything inside a **blue box like this one** is extra material that I added.

Pagination — and why this file is called “Lecture 16” too. These sheets are numbered 19–33, and they continue the chapter that **Lecture 15** belongs to: that chapter ran [p. 1*****]–[p. 18*****] (the eighth restart of the numbering), ending with the extra comment on off-shell gauge invariance. This addendum picks up at [p. 19*****] and runs to [p. 33*****], so the page markers here carry **eight stars** — *not* the nine stars of the gravity chapter.

That is the key to the naming puzzle. The lecture *session* numbered 16 evidently spanned a chapter boundary: it first *finished* the QED/gauge-invariance chapter with the photon soft theorem (these sheets), and then *opened* a brand-new chapter, “Gravity as a QFT”, whose numbering restarts at 1 (the file we typeset as Lecture 16). The scans are split by chapter, not by session, so one session produced two files. Lecture 17 then continues the gravity chapter at [p. 11]. See the blue box at the end of Section 1 for why the two halves of that session belong together *physically*.

The scan ends mid-thought. Sheet [p. 33*****] announces a second comment (“★★ COMMENT: let us also elaborate a bit more on the connection between conserved charges and the Ward identity”) and then stops; the remainder of the sheet, and the following sheet, are blank. Section 11 records the fragment and a blue box sketches what the argument would have been. Everything before that is complete and in order.

Where this fits. Lecture 15 established that gauging a global $U(1)$ forces the photon to couple to a conserved current. This addendum shows the *same* statement emerging from the opposite end — from the behaviour of scattering amplitudes when a very soft photon is emitted — and closes the logical circle: gauge invariance $\Leftrightarrow$ conserved current $\Leftrightarrow$ conservation of total charge. Lecture 17 then repeats this entire construction for the graviton, where the analogous argument yields not charge conservation but the **equivalence principle**.

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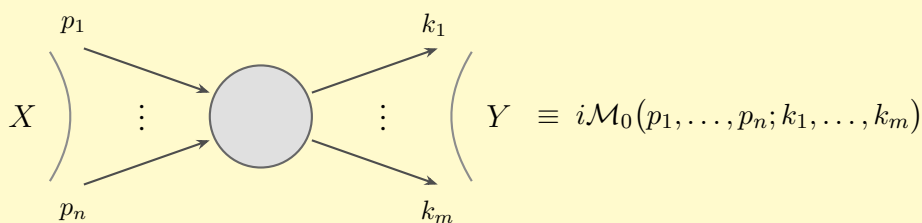
Part I — Setup: the theory and the two classes of diagrams (pp. 19*****–20*****)**1 The Weinberg soft theorem for photons: the setup** [p. 19*****]

We consider the theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \sum_k \left[\bar{\Psi}_k (i\gamma^\mu \partial_\mu - m_k) \Psi_k - q_k A_\mu \bar{\Psi}_k \gamma^\mu \Psi_k \right]$$

↪ The photon interacts with charged particle and antiparticles with masses m_k and charges q_k and $\bar{q}_k = -q_k$.

Consider a scattering process of the kind $X \rightarrow Y$



n particles/antiparticles in the initial state “ X ”

m particles/antiparticles in the final state “ Y ”

↪ We consider the process at some generic order in perturbation theory.

Added explanation (not in the original notes)

What the grey blob means. It is *everything*: all diagrams, at all orders, with any internal structure whatsoever. The whole power of a soft theorem is that we will never need to know what is inside it. Whatever $\mathcal{M}_0$ is, the amplitude with one extra soft photon is $\mathcal{M}_0$ times a universal factor built only from the external momenta and charges.

Why the sum over k in the Lagrangian. We allow *several* species of charged fermion, with independent masses m_k and independent charges q_k . This is essential: the theorem will constrain how those charges are related, and that constraint would be empty if we had assumed a single universal charge from the start. (In Lecture 17 the same care is taken with the κ_i^{IN} , and there the conclusion is far stronger.)

Antiparticles. The single term $-q_k A_\mu \bar{\Psi}_k \gamma^\mu \Psi_k$ describes both particles of charge q_k and antiparticles of charge $-q_k$; this is the field-theoretic origin of antiparticles from Lecture 12, and it is why pp. 24*****–26***** must treat the antiparticle leg separately and will find the sign flip $\bar{q} = -q$ automatically.

Why this material and “Gravity as a QFT” were the same lecture. The photon soft theorem is the last brick of the gauge chapter: it shows that the gauge redundancy of A_μ , seen from the S-matrix side, is conservation of electric charge. The moment that circle closes, the natural question is: what does the same argument give for a massless spin-2 particle? The answer (Lecture 17) is the equivalence principle. Finishing one and starting the other in a single session is exactly the right pedagogical order — and it is why the page numbering of one chapter stops while a new chapter starts at 1.

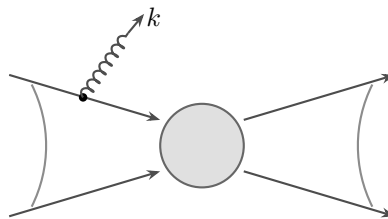
2 Two classes of diagrams, and the soft approximation [p. 20*****]

We now consider, instead of the process $X \rightarrow Y$, the process $X \rightarrow Y + \gamma(k)$, and denote the amplitude for such process

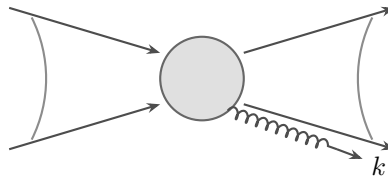
$$i\mathcal{M}_1(p_1, \dots, p_n; k_1, \dots, k_m; k)$$

We separate diagrams contributing to this process into two classes:

i) Photon emission from an external leg — for instance



ii) Photon emission from an internal line — we indicate these diagrams as follows



We compute these diagrams assuming that the photon is “soft” (the energy of the photon is small compared to the characteristic energies and momenta of all other particles involved in the process).

Added explanation (not in the original notes)

The soft limit, quantitatively. “Soft” means $k^\mu \rightarrow 0$, or more precisely $\omega \ll E_i$, $|\vec{p}_i|$ for every hard particle in the process. The emission factors computed below all behave like $1/\omega$, so the amplitude *diverges* as $\omega \rightarrow 0$: this is the infrared divergence of QED, physically the statement that any accelerated charge radiates arbitrarily many arbitrarily soft photons (bremsstrahlung). The divergence cancels in physical cross-sections once one includes virtual soft photons and sums over unresolved final-state radiation — the Bloch–Nordsieck mechanism — but the *coefficient* of the $1/\omega$ pole, which is what we compute here, is exactly what makes the theorem.

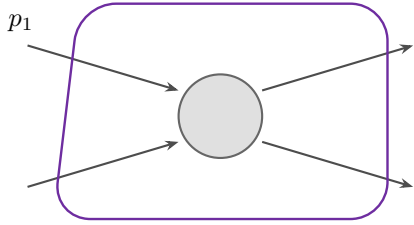
Why external legs dominate. An external leg is on-shell, $p^2 = m^2$; attaching a soft photon puts the adjacent propagator *nearly* on-shell, $(p - k)^2 - m^2 = -2p \cdot k \rightarrow 0$, producing the pole. An internal line is already off-shell by a finite amount, so the extra propagator stays finite and contributes no pole — this is proved on p. 29*****–30*****.

Part II — Emission from external legs

(pp. 21*****–28*****)

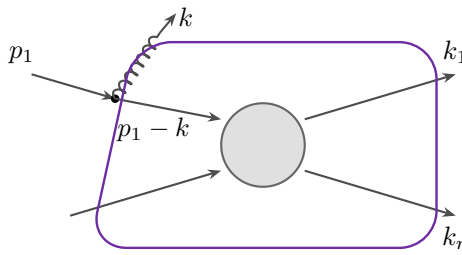
3 Emission from an initial particle leg [p. 21*****–23*****]

We focus on the initial particle leg with momentum p_1 . Consider first, to set our notation, the process without photon emission

$$i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) = \text{Diagram} = i\overline{\mathcal{M}}_0(p_2, \dots, p_n; k_1, \dots, k_m) u(p_1, s_1)$$


We “factor out” from the full amplitude $i\mathcal{M}_0$ the contribution of the initial-state fermion with 4-momentum p_1 .

We consider the diagram with the photon being emitted

$$\text{Diagram} = i\overline{\mathcal{M}}_0(p_1 - k, p_2, \dots, p_n; k_1, \dots, k_m)$$


$\hookrightarrow$ “ q_1^{in} ” is the charge of the incoming particle on leg “1”.

The amplitude is

$$\overline{\mathcal{M}}_0(p_1 - k, \dots, p_n; k_1, \dots, k_m) \underbrace{\frac{i(\not{p}_1 - \not{k} + m_1)}{(p_1 - k)^2 - m_1^2 + i\varepsilon} (-iq_1^{\text{in}} \gamma^\mu) \varepsilon_\mu^*(\vec{k}) u(\vec{p}_1)}_{= \text{“emission factor”}}$$

We focus on the emission factor. We compute it as follows [p. 22*****]

$$\begin{aligned} & \frac{i(\not{p}_1 - \not{k} + m_1)}{\not{p}_1^2 + \not{k}^2 - 2p_1 \cdot k - m_1^2 + i\varepsilon} (-iq_1^{\text{in}}) \gamma^\mu \varepsilon_\mu^*(\vec{k}) u(\vec{p}_1) \quad p_1^2 = m_1^2; \quad k^2 = 0 \\ &= \frac{i(\not{p}_1 - \not{k} + m_1) (-iq_1^{\text{in}}) \gamma^\mu \varepsilon_\mu^*(\vec{k}) u(\vec{p}_1)}{(-2p_1 \cdot k)} \\ &= q_1^{\text{in}} \varepsilon_\mu^*(\vec{k}) \frac{(\not{p}_1 - \not{k} + m_1) \gamma^\mu u(\vec{p}_1)}{-2p_1 \cdot k} \quad (\otimes) \end{aligned}$$

We focus on the numerator $(\not{p}_1 - \not{k} + m_1)\gamma^\mu u(\vec{p}_1)$. We elaborate it as follows

$$\begin{aligned}
 (\not{p}_1 - \not{k} + m_1)\gamma^\mu u(\vec{p}_1) &= \left[(p_1 - k)_\nu \gamma^\nu \gamma^\mu + m_1 \gamma^\mu \right] u(\vec{p}_1) \quad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \\
 &= \left[(p_1 - k)_\nu (-\gamma^\mu \gamma^\nu + 2g^{\mu\nu}) + m_1 \gamma^\mu \right] u(\vec{p}_1) \\
 &= \left[-\gamma^\mu (\not{p}_1 - \not{k}) + 2(p_1 - k)^\mu + m_1 \gamma^\mu \right] u(\vec{p}_1) \\
 &\quad \text{Dirac equation } (\not{p}_1 - m_1)u(\vec{p}_1) = 0 \\
 &= \left[-\gamma^\mu (\not{p}_1 - \not{k}) + 2(p_1 - k)^\mu + m_1 \gamma^\mu \right] u(\vec{p}_1) \\
 &= \left[\gamma^\mu \not{k} + 2(p_1 - k)^\mu \right] u(\vec{p}_1)
 \end{aligned}$$

We thus get, going back to $(*)$ [p. 23*****]

$$\begin{aligned}
 (*) &= q_1^{\text{in}} \varepsilon_\mu(\vec{k})^* \frac{\left[\gamma^\mu \not{k} + 2(p_1 - k)^\mu \right] u(\vec{p}_1)}{-2p_1 \cdot k} \quad \text{transversality } k_\mu \varepsilon^\mu(\vec{k}) = 0 \\
 &= q_1^{\text{in}} \varepsilon_\mu(\vec{k})^* \frac{(\gamma^\mu \not{k} + 2p_1^\mu) u(\vec{p}_1)}{-2p_1 \cdot k} \approx q_1^{\text{in}} \varepsilon_\mu(\vec{k})^* \frac{p_1^\mu u(\vec{p}_1)}{-p_1 \cdot k}
 \end{aligned}$$

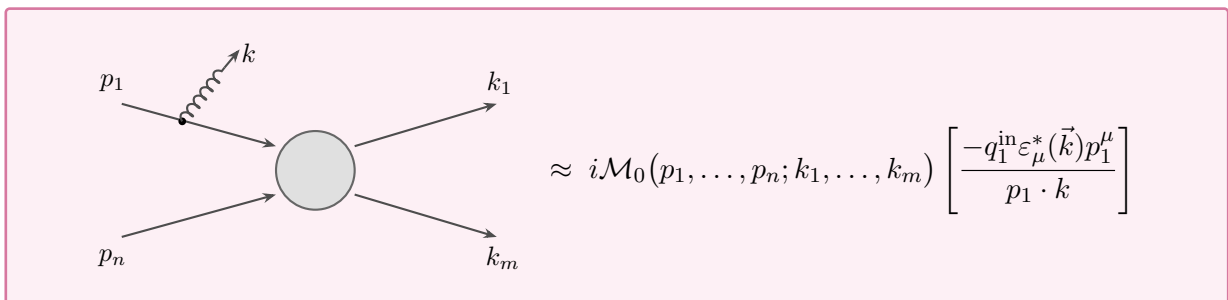
$\hookrightarrow$ We now use the soft approximation, and neglect the k -term at the numerator compared to p_1 .

We go back now to the amplitude of the full diagram

$$\begin{aligned}
 i\overline{\mathcal{M}}_0(p_1 - k, \dots, p_n; k_1, \dots, k_m) \left[\frac{-q_1^{\text{in}} \varepsilon_\mu(\vec{k})^* p_1^\mu}{p_1 \cdot k} \right] u(\vec{p}_1) &= \\
 \approx \underbrace{i\overline{\mathcal{M}}_0(p_1, \dots, p_n; k_1, \dots, k_m) u(\vec{p}_1)}_{\text{it reproduces } i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m)} \left[\frac{-q_1^{\text{in}} \varepsilon_\mu^*(\vec{k}) p_1^\mu}{p_1 \cdot k} \right]
 \end{aligned}$$

In the soft approximation $p_1 - k \approx p_1$

This is a **number**, no spinor structure — all info about the emitted photon factorized



amplitude without soft photon emission

Added explanation (not in the original notes)

The three moves, isolated. The computation looks long but is only three steps.

1. **Denominator:** $(p_1 - k)^2 - m^2 = p_1^2 - 2p_1 \cdot k + k^2 - m^2 = -2p_1 \cdot k$, using $p_1^2 = m_1^2$ (external

leg on-shell) and $k^2 = 0$ (real photon). This is the pole.

2. **Numerator:** anticommute $\not{k}$ and $\not{p}_1$ past γ^μ using $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$, then use the Dirac equation to replace $\not{p}_1 u = m_1 u$. The m_1 terms cancel and what is left is $\gamma^\mu \not{k} + 2(p_1 - k)^\mu$.
3. **Soft limit:** $\gamma^\mu \not{k}$ and k^μ are $O(k)$, while $2p_1^\mu$ is $O(p_1)$ — drop them. What survives has *no gamma matrices left*, so it commutes out of the spinor chain as a plain number.

Why step 3 is the whole point. The emission factor is a c -number, not a matrix: it does not touch the spin structure of the hard process at all. This is what allows it to factorize out in front of $\mathcal{M}_0$ — and it is why the result is independent of the spin of the emitting particle. In Lecture 17 the same statement appears as “independently on the spin of the emitting particle, the denominator of the propagator gives the factor $1/(-2p_i \cdot k)$ ”.

Where transversality is used. $k_\mu \varepsilon^{*\mu} = 0$ kills the $-2k^\mu$ piece of $2(p_1 - k)^\mu$ exactly (not just approximately). The remaining $\gamma^\mu \not{k}$ term is dropped by the soft approximation, being suppressed by one power of ω .

4 Emission from an initial antiparticle leg [p. 24*****-26*****]

We focus on the initial anti-particle leg with momentum p_m . We consider first the process without soft photon emission

$$i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) =$$

$$= \bar{v}(\vec{p}_m) i\widetilde{\mathcal{M}}_0(p_1, \dots, p_n; k_1, \dots, k_m)$$

↪ We factor out the contribution of the incoming antiparticle.

We consider the diagram with the soft photon emission

$$i\widetilde{\mathcal{M}}_0(p_1, \dots, p_m - k; k_1, \dots, k_m)$$

with amplitude given by

$$\bar{v}(\vec{p}_m) (-iq_m^{\text{in}} \gamma^\mu \varepsilon_\mu^*(\vec{k})) \frac{i(-\not{p}_m + \not{k} + m_m)}{(-p_m + k)^2 - m_m^2 + i\varepsilon} i\widetilde{\mathcal{M}}_0(p_1, \dots, p_m - k; k_1, \dots, k_m)$$

We focus on the emission factor, that we compute as follows [p. 25*****]

$$\bar{v}(\vec{p}_m) q_m^{\text{in}} \gamma^\mu \varepsilon_\mu^*(\vec{k}) \frac{(-\not{p}_m + \not{k} + m_m)}{\not{p}_m^2 + \not{k}^2 - 2p_m \cdot k - m_m^2} = \bar{v}(\vec{p}_m) \frac{\gamma^\mu (-\not{p}_m + \not{k} + m_m)}{-2p_m \cdot k} q_m^{\text{in}} \varepsilon_\mu^*(\vec{k}) \quad (\otimes)$$

We elaborate the numerator as follows

$$\begin{aligned} \bar{v}(\vec{p}_m) \gamma^\mu (-\not{p}_m + \not{k} + m_m) &= \bar{v}(\vec{p}_m) \left[\gamma^\mu \gamma^\nu (-p_m + k)_\nu + \gamma^\mu m_m \right] \\ &= \bar{v}(\vec{p}_m) \left[(-\gamma^\nu \gamma^\mu + 2g^{\mu\nu}) (-p_m + k)_\nu + \gamma^\mu m_m \right] \\ &= \bar{v}(\vec{p}_m) \left[(\not{p}_m - \not{k}) \gamma^\mu + 2(-p_m + k)^\mu + \gamma^\mu m_m \right] \\ &= \bar{v}(\vec{p}_m) \left[-\not{k} \gamma^\mu + 2(k - p_m)^\mu \right] \end{aligned}$$

using Dirac equation $\bar{v}(\vec{p}_m) (\not{p}_m + m_m) = 0$

$$\begin{aligned} (\otimes) &= \bar{v}(\vec{p}_m) \frac{[-\not{k} \gamma^\mu + 2(k - p_m)^\mu]}{-2p_m \cdot k} q_m^{\text{in}} \varepsilon_\mu^*(\vec{k}) = \bar{v}(\vec{p}_m) \frac{(-\not{k} \gamma^\mu - 2p_m^\mu)}{-2p_m \cdot k} q_m^{\text{in}} \varepsilon_\mu^*(\vec{k}) \\ &\approx \bar{v}(\vec{p}_m) \left[\frac{q_m^{\text{in}} p_m^\mu \varepsilon_\mu^*(\vec{k})}{p_m \cdot k} \right] \quad \text{soft approximation} \end{aligned}$$

The amplitude of the full diagram is

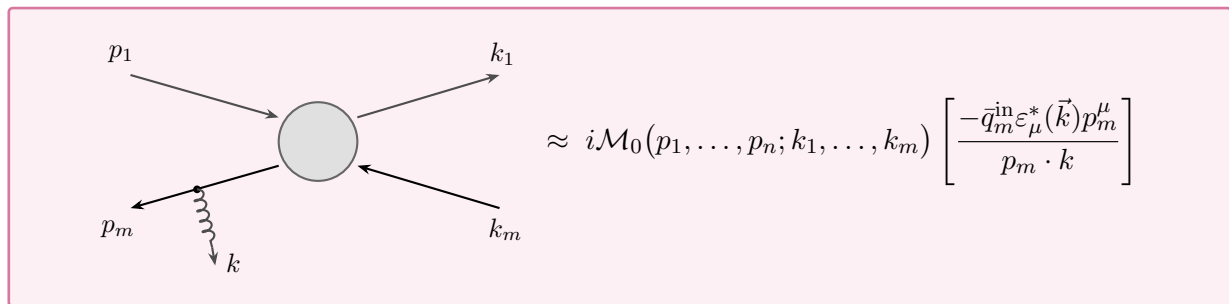
$$\bar{v}(\vec{p}_m) \left[\frac{q_m^{\text{in}} p_m^\mu \varepsilon_\mu^*(\vec{k})}{p_m \cdot k} \right] i\widetilde{\mathcal{M}}_0(p_1, \dots, \underbrace{p_m - k}_{\approx p_m}; k_1, \dots, k_m) =$$

[p. 26*****]

$$= \left[(-q_m^{\text{in}}) \frac{\varepsilon_\mu^*(\vec{k}) p_m^\mu}{-p_m \cdot k} \right] \underbrace{\bar{v}(\vec{p}_m) i\widetilde{\mathcal{M}}_0(p_1, \dots, p_m; k_1, \dots, k_m)}_{= i\mathcal{M}_0(p_1, \dots, p_m; k_1, \dots, k_m)}$$

This is $\bar{q}_m^{\text{in}} = -q_m^{\text{in}}$

We thus get



↪ The form of this factor is always the same, it has the structure

$$-Q_j^{\text{in}} \frac{\varepsilon_\mu^*(\vec{k}) p_j^\mu}{p_j \cdot k}$$

where Q_j^{in} is the **electric charge of the emitting particle**

$Q_j^{\text{in}} \equiv +q_j^{\text{in}}$ for an emitting particle; $Q_j^{\text{in}} = -q_j^{\text{in}}$ for an emitting anti-particle

Added explanation (not in the original notes)

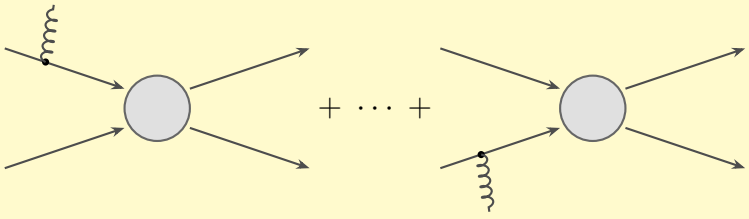
What was gained by redoing the computation for an antiparticle. Structurally nothing changes — same pole, same gamma-matrix algebra, same soft truncation. What changes is *bookkeeping*: the spinor is $\bar{v}$ instead of u , it sits to the *left* of the vertex, the Dirac equation used is $\bar{v}(\not{p} + m) = 0$ (with a $+$ sign), and the propagator momentum is $-p_m + k$. The net effect is one overall sign, which is precisely the statement that the antiparticle carries charge $-q$.

So the single symbol Q_j does all the work. From here on the notes never again distinguish particles from antiparticles: the emission factor is $-Q_j \varepsilon_\mu^* \cdot p_j / (p_j \cdot k)$ with Q_j the *physical electric charge* of whatever sits on leg j . This is exactly the unification that Lecture 12 predicted — one field, two species, opposite charges.

5 Sum over initial-state emissions, and final-state legs [p. 27*****–28*****]

We thus collect all diagrams with emission from an initial-state leg as follows [p. 27*****]

Sum over initial-state emissions

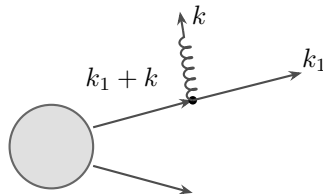


$$\approx i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) \sum_{j=1}^n (-1) Q_j^{\text{in}} \frac{\varepsilon_\mu^*(\vec{k}) p_j^\mu}{p_j \cdot k}$$

■ Emission from final state legs

The result is very similar to the previous one, with the only difference given by a **change of sign**

↪ Consider, for instance, emission from a final state particle, and focus on the emission factor



$$\begin{aligned} &= \bar{u}(\vec{k}_1) (-iq_1^{\text{FIN}} \gamma^\mu \varepsilon_\mu^*(\vec{k})) \frac{i(\not{k}_1 + \not{k} + m_1)}{(k_1 + k)^2 - m_1^2 + i\varepsilon} = \\ &= \bar{u}(\vec{k}_1) q_1^{\text{FIN}} \gamma^\mu \varepsilon_\mu^*(\vec{k}) \frac{(\not{k}_1 + \not{k} + m_1)}{\cancel{\not{k}_1^2} + \cancel{\not{k}^2} + 2k_1 \cdot k - \cancel{m_1^2} + i\varepsilon} = \\ &= q_1^{\text{FIN}} \varepsilon_\mu^*(\vec{k}) \bar{u}(\vec{k}_1) \frac{\gamma^\mu (\not{k}_1 + \not{k} + m_1)}{2k_1 \cdot k} \quad (\otimes) \end{aligned}$$

As usual [p. 28*****]

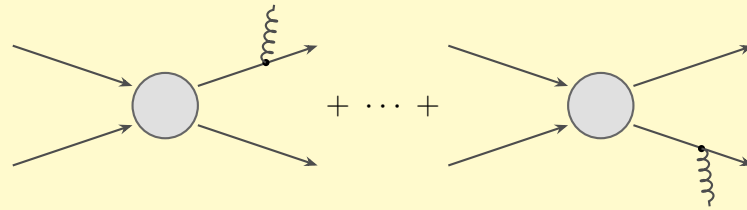
$$\begin{aligned} \bar{u}(\vec{k}_1) \gamma^\mu [(k_1 + k)_\nu \gamma^\nu + m_1] &= \bar{u}(\vec{k}_1) [\gamma^\mu \gamma^\nu (k_1 + k)_\nu + \gamma^\mu m_1] \\ &= \bar{u}(\vec{k}_1) [(-\gamma^\nu \gamma^\mu + 2g^{\mu\nu})(k_1 + k)_\nu + \gamma^\mu m_1] \\ &= \bar{u}(\vec{k}_1) [-\cancel{\not{k}_1} + \not{k} \gamma^\mu + 2(k_1 + k)^\mu + \cancel{\gamma^\mu m_1}] \\ &= \bar{u}(\vec{k}_1) [-\not{k} \gamma^\mu + 2(k_1 + k)^\mu] \end{aligned}$$

And

$$(\otimes) = \bar{u}(\vec{k}_1) q_1^{\text{FIN}} \varepsilon_\mu^*(\vec{k}) \frac{(-\not{k} \gamma^\mu + 2k_1^\mu + 2k^\mu)}{2k_1 \cdot k} \approx \left[\frac{q_1^{\text{FIN}} \varepsilon_\mu^*(\vec{k}) k_1^\mu}{k_1 \cdot k} \right] \bar{u}(\vec{k}_1)$$

opposite sign compared to the emission from the initial state particle

We thus write the result



$$\approx i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) \sum_{j=1}^m Q_j^{\text{FIN}} \frac{\varepsilon_\mu^*(\vec{k}) k_j^\mu}{k_j \cdot k}$$

Added explanation (not in the original notes)

Where the relative sign really comes from. Only one thing differs between the two cases: the momentum flowing in the internal propagator. Incoming leg: $p_i - k$, giving $(p_i - k)^2 - m^2 = -2p_i \cdot k$. Outgoing leg: $k_j + k$, giving $(k_j + k)^2 - m^2 = +2k_j \cdot k$. Hence the minus sign between the two sums — exactly the same bookkeeping as on p. 26^{*****} of Lecture 17 for gravitons.

Physical reading. The initial-state and final-state radiation interfere with opposite signs. This is why a charge that is not deflected radiates nothing: if $p_i = k_j$ and $Q_i = Q_j$, the two terms cancel identically. Radiation requires acceleration — the soft theorem knows classical electrodynamics.

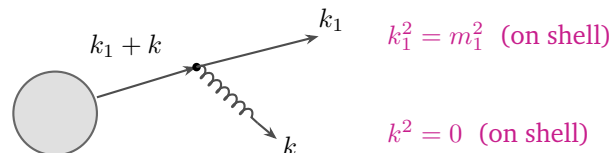
Part III — The soft theorem and the closing of the circle (pp. 29*****–33*****)

6 Emission from internal lines is sub-leading [p. 29*****–30*****]

The claim is that, in the soft limit, all these diagrams give a sub-leading contribution to the amplitude if compared to the emission from external legs.

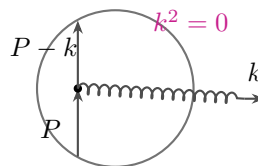
The argument goes as follows.

↪ If we emit from an external leg



↪ The intermediate propagator develops a behavior $\frac{1}{k_1 \cdot k}$ in the limit of soft “ k ”.

↪ If we emit from an internal line, we have the following situation



↪ Consider one of the two propagators. We have:

$$\frac{1}{(P - k)^2 - m^2} = \frac{1}{P^2 + k^2 - 2P \cdot k - m^2} = \frac{1}{P^2 - 2P \cdot k - m^2} \underset{\text{soft limit}}{\approx} \frac{1}{P^2 - m^2}, \quad P^2 \neq m^2$$

[p. 30*****]

↪ Since the internal line of momentum P is not on-shell ($P^2 \neq m^2$), in the soft limit we do not have a factor $\frac{1}{P \cdot k}$ but a factor

$$\frac{1}{P^2 - m^2} \ll \frac{1}{P \cdot k},$$

much smaller than those arising from external legs.

Added explanation (not in the original notes)

The one-line version. On an external leg the “ $-m^2$ ” cancels the “ P^2 ” because the leg is on-shell, so the whole denominator is $O(k)$ and the propagator blows up like $1/\omega$. On an internal line $P^2 - m^2$ is a finite non-zero number, so the $-2P \cdot k$ correction is a small perturbation and the propagator stays finite. One diagram class has a pole, the other does not — and in the limit $\omega \rightarrow 0$ the pole wins by an arbitrarily large factor.

Consequence. All the model-dependence of the emission is confined to the non-singular part. The *leading* soft behaviour is universal, determined entirely by the external data $\{Q_j, p_j\}$. This is what turns a statement about Feynman diagrams into a theorem.

7 The full amplitude and the soft theorem [p. 30*****–31*****]

Let us write the full amplitude in the soft limit

$$i\mathcal{M}_1(p_1, \dots, p_n; k_1, \dots, k_m; k) \approx i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) \varepsilon_\mu^*(\vec{k}) \times$$

$$\underbrace{\left[\sum_{j=1}^m Q_j^{\text{FIN}} \frac{k_j^\mu}{k_j \cdot k} - \sum_{i=1}^n Q_i^{\text{IN}} \frac{p_i^\mu}{p_i \cdot k} \right]}_{\text{soft emission factor}}$$

Now the theorem. We know that if we substitute $\varepsilon_\mu(\vec{k})^* \rightarrow k_\mu$, we must get zero (**Ward identity**). [p. 31*****]

Given the amplitude we wrote before, we must have

$$i\mathcal{M}_0 k_\mu \left(\sum_j Q_j^{\text{FIN}} \frac{k_j^\mu}{k_j \cdot k} - \sum_i Q_i^{\text{IN}} \frac{p_i^\mu}{p_i \cdot k} \right) \stackrel{!}{=} 0, \quad \text{which is}$$

$$i\mathcal{M}_0 \left(\sum_j Q_j^{\text{FIN}} \frac{\cancel{k} \cdot \cancel{k}_j}{\cancel{k}_j \cdot \cancel{k}} - \sum_i Q_i^{\text{IN}} \frac{\cancel{p}_i \cdot \cancel{k}}{\cancel{p}_i \cdot \cancel{k}} \right) \stackrel{!}{=} 0$$

$$\sum_{j=1}^m Q_j^{\text{FIN}} = \sum_{i=1}^n Q_i^{\text{IN}}$$

conservation of total charge

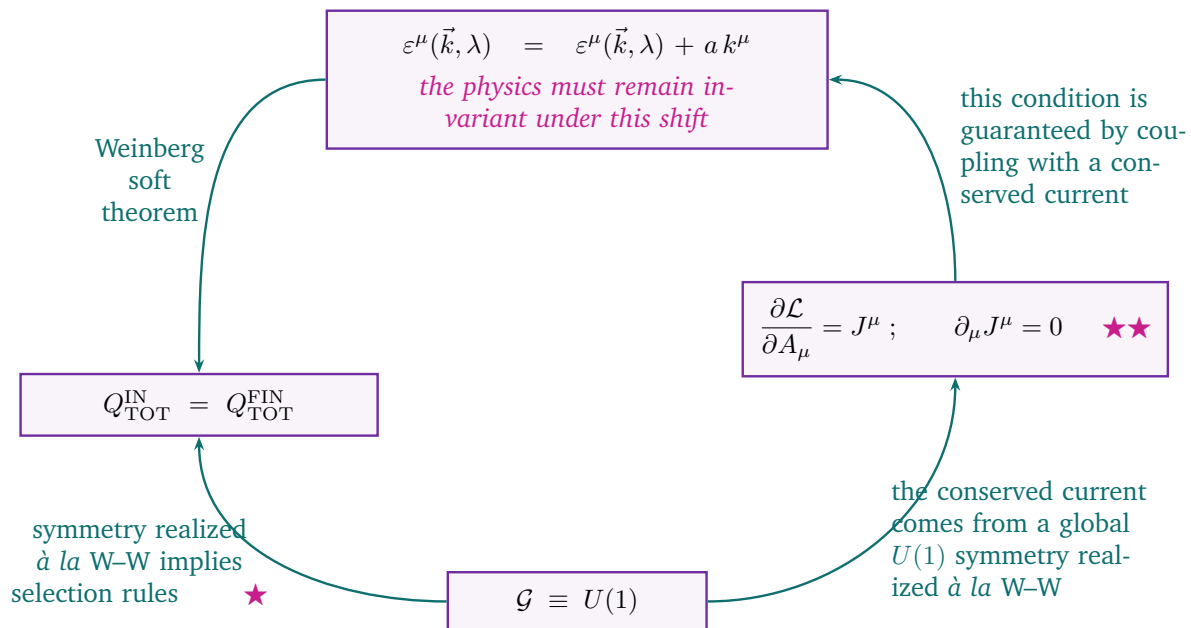
Added explanation (not in the original notes)

What has just happened. We never assumed charge conservation. We assumed only that the photon is a massless spin-1 particle whose polarization is defined up to $\varepsilon^\mu \rightarrow \varepsilon^\mu + a k^\mu$ (gauge invariance), and we computed the universal soft emission factor. Demanding that the gauge shift not change the amplitude then *forces* the sum of incoming charges to equal the sum of outgoing charges — in *every* process, at *every* order.

Compare with Lecture 17. There the same manipulation gave $\sum_j \kappa_j^{\text{FIN}} k_j^\mu = \sum_i \kappa_i^{\text{IN}} p_i^\mu$, an equation between four-vectors rather than numbers, which combined with momentum conservation to force all κ 's *equal*. The difference is the single extra power of momentum in the graviton emission factor $\varepsilon_{\mu\nu}^* p^\mu p^\nu$. Spin 1 gives a conservation law; spin 2 gives universality.

8 The “closing of the circle” [p. 31*****]

We arrive at the following “closing of the circle”



Added explanation (not in the original notes)

How to read the diagram. Four statements, each implying the next, arranged in a loop:

1. *Top:* the photon polarization is physically defined only up to $a k^\mu$ — gauge redundancy at the level of one-particle states.
2. *Left:* applied to the soft emission factor, that redundancy yields $Q_{\text{TOT}}^{\text{IN}} = Q_{\text{TOT}}^{\text{FIN}}$ — the Weinberg soft theorem.
3. *Bottom:* the same selection rule follows independently from having an unbroken (Wigner–Weyl) global $U(1)$ symmetry, whose Noether charge labels the asymptotic states. This is comment ★, proved on pp. 32*****–33*****.
4. *Right:* that global $U(1)$ supplies a conserved current J^μ , and coupling A_μ to it is exactly what guarantees the gauge invariance we started from (Lecture 15).

The circle closes: *gauge redundancy* $\Leftrightarrow$ *conserved current* $\Leftrightarrow$ *unbroken $U(1)$* $\Leftrightarrow$ *charge conservation*. The same physical fact, reached from the Lagrangian, from the symmetry algebra, and from the S-matrix.

“À la Wigner–Weyl” means the symmetry is realized *unbroken*: the vacuum is invariant, $Q|0\rangle = 0$, and states organize into eigenstates of Q . The alternative — Nambu–Goldstone realization, where the vacuum is not invariant — would destroy the selection rule and give massless Goldstone bosons instead. That distinction is what the phrase is flagging.

9 Comment ★: symmetries $\rightarrow$ selection rules [p. 32*****–33*****]

Let us elaborate a bit more about this point (symmetries $\rightarrow$ selection rules).

Consider a scattering process

$$S_{fi} = \langle f | S | i \rangle$$

multiparticle free states

$$\hookrightarrow S = \sum_{m=0}^{\infty} \frac{(-i)^m}{m!} \int d^4x_1 \cdots d^4x_m T[\mathcal{H}_{\text{int}}(x_1) \cdots \mathcal{H}_{\text{int}}(x_m)]$$

In the presence of a $U(1)$ symmetry realized *à la* W–W, we know that states are organized in such a way that they are eigenstates of the generator Q (the Noether charge of the $U(1)$ symmetry)

$$\begin{aligned} Q|i\rangle &= Q_{\text{TOT}}^{\text{IN}}|i\rangle \\ Q|f\rangle &= Q_{\text{TOT}}^{\text{FIN}}|f\rangle \end{aligned}$$

And, consequently,

$$\begin{aligned} \exp(-iQ\theta)|i\rangle &= e^{-i\theta Q_{\text{TOT}}^{\text{IN}}}|i\rangle \\ \langle f|\exp(iQ\theta) &= \langle f|e^{+i\theta Q_{\text{TOT}}^{\text{FIN}}} \end{aligned}$$

$$\exp(-iQ\theta) \equiv U(\theta)$$

We thus write [p. 33*****]

$$S_{fi} = \langle f|S|i\rangle = \langle f|U(\theta)^\dagger \underbrace{U(\theta) S U(\theta)^\dagger}_{} U(\theta)|i\rangle =$$

By construction, the Lagrangian, and thus $\mathcal{H}_{\text{int}}(x)$, is invariant under global symmetry transformations: $U(\theta) S U(\theta)^{-1} = S$

$$= e^{i\theta(Q_{\text{TOT}}^{\text{FIN}} - Q_{\text{TOT}}^{\text{IN}})} \langle f|S|i\rangle \quad \Longrightarrow \quad \boxed{Q_{\text{TOT}}^{\text{FIN}} = Q_{\text{TOT}}^{\text{IN}}}$$

which is the selection rule found before.

Added explanation (not in the original notes)

The argument in one sentence. If S commutes with Q , then $\langle f|S|i\rangle$ picks up a phase $e^{i\theta(Q_{\text{TOT}}^{\text{FIN}} - Q_{\text{TOT}}^{\text{IN}})}$ under the symmetry transformation — but it must be *unchanged*, so either the phase is trivial or the amplitude vanishes. Hence: *any process with $Q_{\text{TOT}}^{\text{FIN}} \neq Q_{\text{TOT}}^{\text{IN}}$ has exactly zero amplitude.* That is what a selection rule is.

Why this is the same result by a completely different route. The soft-theorem derivation (p. 31*****) used dynamics: propagators, vertices, gauge invariance of amplitudes. This derivation uses only the algebra of states and the invariance of S . Neither knows about the other, and they agree — which is the “closing of the circle”.

A subtlety worth noting. The step $USU^\dagger = S$ requires the interaction Hamiltonian to be invariant *and* the vacuum (hence the asymptotic states’ construction) to be invariant — i.e. the Wigner–Weyl realization again. If the $U(1)$ were spontaneously broken, $Q|0\rangle \neq 0$, the states would not be charge eigenstates and the selection rule would fail.

10 Comment ★★ [p. 33*****]

Let us also elaborate a bit more on the connection between conserved charges and the Ward identity...

[The sheet stops here; the rest of the page, and the following sheet, are blank.]

Added explanation (not in the original notes)

What this comment would have said. The missing argument is the operator derivation of the Ward–Takahashi identity from current conservation. In outline: insert the conserved current into a Green’s function,

$$\partial_\mu \langle 0 | T \{ J^\mu(x) \Psi(y) \bar{\Psi}(z) \} | 0 \rangle = -i\delta^4(x-y) \langle \dots \rangle + i\delta^4(x-z) \langle \dots \rangle,$$

where the contact terms on the right come from differentiating the time-ordering θ -functions and using the equal-time commutator $[J^0(x), \Psi(y)]_{x^0=y^0} = -q \delta^3(\vec{x} - \vec{y}) \Psi(y)$ — i.e. from the statement that Q generates the symmetry. In momentum space this becomes $k_\mu \mathcal{M}^\mu = 0$ for on-shell external states: **exactly the Ward identity used on p. 30*******.

So the logic completes as follows: *conserved charge* $\Rightarrow$ *equal-time commutator* $\Rightarrow$ *Ward identity* $\Rightarrow$ *decoupling of the gauge shift* $\varepsilon^\mu \rightarrow \varepsilon^\mu + a k^\mu$. The arrow on the right-hand side of the “closing of the circle” diagram is thereby justified from first principles rather than asserted. This is a standard textbook derivation (Peskin & Schroeder §7.4), so nothing essential is lost by the scan ending here.

Summary — the logic in seven lines

1. Take QED with any number of charged species and any hard process $X \rightarrow Y$ with amplitude $\mathcal{M}_0$; add one soft photon.
2. Diagrams split into emission from external legs and from internal lines.
3. An external leg is on-shell, so the adjacent propagator gives a pole $1/(\mp 2p \cdot k)$; the gamma-matrix algebra plus the Dirac equation reduce the numerator to $2p^\mu$, leaving a pure number with no spinor structure.
4. Internal-line emission has no pole and is sub-leading.
5. Summing external emissions, $\mathcal{M}_1 \approx \mathcal{M}_0 \varepsilon_\mu^* [\sum_j Q_j^{\text{FIN}} k_j^\mu / (k_j \cdot k) - \sum_i Q_i^{\text{IN}} p_i^\mu / (p_i \cdot k)]$, with $Q = \pm q$ for particles/antiparticles.
6. The Ward identity ($\varepsilon_\mu^* \rightarrow k_\mu$ must annihilate the amplitude) then forces $\sum_j Q_j^{\text{FIN}} = \sum_i Q_i^{\text{IN}}$: **charge conservation**.
7. The same selection rule follows independently from an unbroken (Wigner–Weyl) global $U(1)$ acting on the asymptotic states — closing the circle with Lecture 15’s gauge construction.

And the sequel. Replace the photon by a massless spin-2 particle: the emission factor becomes $\varepsilon_{\mu\nu}^* p^\mu p^\nu / (p \cdot k)$, and the identical argument gives not a conservation law but the *equality of all couplings* — the equivalence principle. That is Lecture 17.